Do AI-Enabled Website Builders Dream of Dark-Patterned Sheep? Understanding AI-Enabled Design Tools

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Abstract

Rapid website deployment facilitated by AI-enabled design tools may exacerbate the problem of dark patterns in the open internet, especially for sites created by non-designers or non-developers who may lack the technical or practical expertise in designing effective but user-protective websites. We refer to such website creators as potential "deployers" of dark patterns. This research statement explores the intersection of dark patterns and generative AI scholarship, discussing how generative AI website design tools might enable the proliferation of dark patterns made by unaware deployers. We discuss our preliminary exploration of generative AI app development tools – investigating how such tools may either be part of the problem or solution depending on factors of their use, then outline our research agenda and key questions for the workshop community.

CCS Concepts

• **Human-centered computing → Systems and tools for interaction design**; *Empirical studies in HCI*; • **Social and professional topics → Computing / technology policy**.

Keywords

dark patterns, technology compliance, design tools, user experiences, website design

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1 Introduction

Dark patterns[1] , often described as design elements that interfere with user behavior against the user's best interests or intended actions, are well-established in both scholarly literature [11, 20] and regulation or enforcement [1, 9, 24]. Scholarship and industry alike have formed robust taxonomies [3, 9, 10, 23] and ontologies [11] or theoretical work [4] to characterize and structure the concept of dark patterns, providing guidance for identifying dark patterns in common interfaces. User studies examine user awareness or impact [2, 21, 25, 28, 29], the viability of interventions [7, 18, 25], and impact on different users [22, 25]. Other scholarship explores dark patterns in-the-wild across user experiences in differing modalities [12, 16, 17, 19, 21, 35], platform types [13, 15, 29].

Methods for dark patterns *deployment*, however, are comparatively under-studied. Designs may be constructed by experienced practitioners with formal training in user experience (UX) or web design, but websites are increasingly being designed via assisted coding tools powered by large language models (LLMs) or AI-enabled website builders. Such models are based on statistical patterns in large datasets, and may thus have been trained on websites that include dark patterns. Here, we investigate whether and when coding models autonomously introduce dark patterns when prompted to build websites, and the types of prompts and models that affect the results. Such a tendency may lead to an increase of dark patterns in practice, especially with non-designer deployers who may not be able to detect and remedy these issues. Beyond reducing trust and introducing friction to the internet, deployers may face legal risk, given dark patterns regulation like the EU Digital Services Act (DSA) [24] and anti-deception provisions in the EU AI Act [8].

This study specifically focuses on the AI-enabled tools available to a range of dark pattern or design deployers at varying technical or UX skill levels. Specifically, we are investigating the following research questions:

- **RQ1:** *Do the resultant dark pattern experiences of websites differ for sites constructed by AI design tools of varying sophistication?* One-stop-shop interface builders like website creation or app development platforms now offer AI functionality, as do lower-level design or programming tools (e.g., Cursor). This diversity captures a wide variety of deployer skill level.
- **RQ2:** *How do the resultant sites compare when instructing AI tools to avoid dark patterns for compliance?* We investigate

[1] Dark pattern practices may be referred to by alternative names, such as "deceptive design practices", "manipulative design", and other subsets like "attention capture design patterns (ACDPs)". To encompass this variety, we retain "dark patterns" for the sake of this work insofar as their designs inherently subvert user expectation or keep users "in the dark", but acknowledge and encourage the ongoing debate regarding the potential racial connotations of the term "darkness".

whether adjusted prompting or generative AI use consistently decreases the prevalence of dark patterns across these various tools.
- **RQ3:** *How do the design deployment experiences differ across tools, in the effort to build sites with minimal dark patterns?* We question whether more user-friendly tools result in a more fluid experience for deploying more compliant interfaces.

To investigate these questions, we aim to examine AI-enabled website creation tools across a range of technical capacity and complexity, enumerating and documenting their features to establish a baseline of what features they share: Cursor, Github Copilot, Squarespace, and equivalents. To this end, we generate e-commerce websites with popular site-building tools, across three different test cases for intended outcomes. This workshop research statement articulates our preliminary tests using Github Copilot to create an initial corpus of AI-generated 7 websites, to help inform future experiments on the broader expanse of site-building tools. The tests are performed from the perspective of a non-technical user who provides high-level instructions to an LLMs to build a website.

2 Dark Patterns and Generative AI

The intersection of dark patterns and generative AI (and particularly LLMs) involves non-exhaustive topics: dark patterns as discovered within LLMs as a modality (e.g., deceptive design in chatbots), LLMs to create deceptive interfaces, and LLMs to detect deceptive designs. Shi et al. [31] explore the possibility of dark patterns *within* conversational LLM interactions, an emergent topic of interest in the wake of high-profile cases highlighting LLMs' potential harms [14].

For the latter category of work, some scholars have recently started to test the role of AI in dark pattern generation, particularly exploring whether dark patterns might arise as a byproduct of the websites used in a model's training data. Serezlic and Quijada [30] utilize the Gray et al. [11] ontology in a participatory study investigating how generative AI might build sites that avoid dark patterns when explicitly instructed, also inspecting whether dark patterns remain despite specific prompting against their inclusion. Similarly, Xu and Al-Mashahedi [37] conduct an A/B test experiment with eye tracking data to understand how users react to human or AI-built dark patterns, finding AI-enhanced dark patterns more effective against users than human-designed patterns.

LLMs for Dark Pattern Detection. A growing body of work explores how dark patterns might be identified with LLMs [5, 6] or how LLMs may be used to agentically avoid dark patterns [34]. Similarly, Schäfer et al. [27] explore the use of LLMs to effectively re-design extant dark designs, using LLMs to not only identify dark patterns but to rectify their effect towards a reduction in manipulation. Our study departs from the precedent set in Serezlic and Quijada [30], Xu and Al-Mashahedi [37], which explore whether LLMs facilitate dark pattern avoidance. That is, we do not conduct participant studies to evaluate the AI-enabled website tools. Our methods relate to Schäfer et al. [27] insofar as we explore how LLMs may be used to bring interfaces towards minimized deception or manipulation, but rather than examine extant interfaces, we examine the process of site-building from scratch. That is, while Schäfer et al. [27] demonstrate that LLMs can be used to minimize dark patterns, we expand our study to examine how deployers as users should leverage such tools to overcome skills gaps.

AI Design and Development Tools. Popular website design tools like Shopify, Wix, and Squarespace have joined countless technology companies in incorporating "AI" functionality in their existing products [32, 33, 36]. These product or service landing pages tout taglines and pitches like *"Create a custom website with AI/Generate a personalized storefront with our built-in, AI-powered functionality for commerce"* [32] or *"Experience site creation like never before/Get a unique, business-ready website in no time with our powerful AI"* [36].

Design Regulation in the Age of AI. At present, a regulatory momentum has begun to more closely regulate designs, particularly those pertaining to consent mechanisms or privacy options. The EU Digital Services Act (DSA) famously includes explicit references and definitions of dark patterns in Recital 67 [24], while ample dark patterns scholarship has been oriented around the EU General Data Protection Regulation (GDPR) requirements for consent and choice for user privacy.

However, the DSA's concern with primarily very large online platforms (VLOPs)[2] leaves an important gap; VLOPs are more likely to have large development teams with hired UX expertise, whereas one-off sites built by individuals or smaller tech teams with website building services or AI coding tools may slip through liability cracks. Compliance (and the resulting, hopefully-less-deceptive designs therein) thus might become a "problem" only for VLOPs who may have the capacity to fix design problems. Smaller firms, or the deployers our study is concerned with, may never have the resources for effective and total compliance to often-intricate laws by virtue of their size or accesible expertise. In response, compliance or detection tools as mentioned in earlier in this section might be called upon by smaller deployers in order to comply.

3 Methods and Proposed Work

To establish a baseline of common features or capabilities across the tools examined, we first conduct a pilot exploration of Github Copilot. This familiarises our team with a low-code but potentially sophisticated tool, and additionally facilitates the selection of other tools for comparative analysis in future work, e.g., high-code solutions like Cursor or WYSIWYG builders like Squarespace and Wix. Our planned work and research direction beyond this preliminary experiment is further articulated in § 4.

3.1 Test Case Design

Following the pilot study, we then drafted a set of requirements for a website for each tool to construct, with varying levels of human intervention. We chose cases that would either be popular uses for such tools, as well as those that have been identified in prior literature to often contain dark patterns. Specifically, we sought to use each tool to build a simple e-commerce website (e.g., for a user who might be an independent vendor looking to sell products or services). We build these prompts to mimic the behavior of a deployer with limited knowledge of website design or software development. For the purpose of our preliminary Github Copilot tests, we assume limited knowledge of software development toolkits,

[2] Specifically, the DSA regulates platforms with over 45 million users in the EU.

but that the deployer knows about Github and minimally how to create issues for Copilot on to build their desired website.

Specifically, we investigate three initial test cases:

- **C1 (Baseline website)**: build the requested site without additional compliance constraints or direction
- **C2 (Provide design guidance)**: build the site while avoiding explicitly named dark patterns
- **C3 (Provide regulatory guidance)**: build the requested site to comply with the EU DSA and AI Act

C1's default website creation involves attempting to build a website as agnostic of bright or dark patterns – or otherwise as minimally functional – as possible. For generative AI tools, particularly low-code application development, we use basic, minimized prompts, e.g., "build an e-commerce site with XYZ features" without instructing the tool to optimize for particular goals (such as engagement, sales turnover, etc.). For **C2**, we attempt to build the website in a manner that tries to avoid dark patterns and related behaviors at a high level. For genAI tools, our prompts now additionally request the tool to "avoid incorporating dark pattern design elements" atop the extant C1 prompt, and explicitly target a subset of five dark patterns to test the genAI performance.[3] In WYSIWYG tools, we will select options within any AI-enabled workflow provided by the platform that minimize dark patterns or avoid options that deploy them. Finally, for **C3**, we seek to build sites that are DSA/AI Act-compliant – that is, we expressly avoid any options in WYSIWYG editors that implement behaviors suiting regulatory requirements for dark patterns, and with LLMs prompt for compliant sites.

For each case category, we create two websites with prompts of differing levels of detail. The first simply prompts Copilot to create an e-commerce site according to the case requirements, while the second iteration prompts Copilot to create an e-commerce site that must have (1) registration components, (2) a cookie banner, and (3) account deletion features. We intend to conduct each test on every website builder tool in our final study, but for preliminary exploration focus exclusively on creating these websites via one medium (Github Copilot) to start. Thus, for each case category, we will follow tools' instructions to build sites towards each case and sub-objectives.

Building Upon Prior Work. The pilot methods used in this early position paper are strongly aligned with Schäfer et al. [27], which improves the designs of extant websites with GPT-4o to expand beyond simple DP detection and instead *mitigate* or otherwise minimize dark patterns in-the-wild. [27]'s approach improves the same set of initial websites iteratively, whereas our similar prompting instead creates these websites afresh for direct comparison across all six test cases as newly-created websites, but we do believe that a separate study explicitly using [27]'s methods on websites build entirely with AI-enabled tools would be an interesting and fruitful area for future work. Naturally, our initial results (which at the moment only pertain to fully-LLM-built websites) appear at a high-level to align with the [27]'s results, but we expect to be able to better contextualize our findings with prior work after refining our methods for the full study. We perceive our proposed study as complementary to Schäfer et al. [27], and hope to expand upon implications for from-scratch development (and the development experience) in our work.

[3]In this pilot test, we select four Interface Interference and one Social Engineering meso-level Gray et al. [11] ontology dark patterns based on what we expect to find on a simple e-commerce site, but in the full study will expand this set with further refinement after additional testing.

3.2 Site Generation

GPT-5.2-Codex was utilized to generate the websites for various zero-shot prompts. We chose zero-shot prompts to begin with to see how Copilot might interpret user requests with minimal direction. In a new empty repository, we created Github issues per-prompt, for example: *"Create an e-commerce site. It must have the following features: <listed features>"*, with or without details based on the prompt type. Prompts were designed to instruct Copilot to create a website, and in this preliminary study we elected to build e-commerce sites to more closely align with the types of sites a layperson deployer might be likely to build. We then assigned the issue to Copilot. Copilot then executed a pull request to deliver the requested website on a fresh branch in our repository, which we then pulled locally and serve via `localhost` to evaluate the generated websites. No external resources or information was provided. Copilot agents ran in the background, and indicated work completion.

Specifications for each website built, mappings to each case, and the site keys are provided in Table 1. We generated these websites between December 2025-January 2026.

Documenting AI-Enabled Tool Experiences. Though this stage of the work only produces the LLM portion of our intended study, to better understand how websites are deployed and the experiences of potential deployers, we are developing a standardized workflow script as noted above, then intend to screen-record our interactions with WYSIWYG tools. Qualitative content analysis may then be used to evaluate and characterize similarities, differences, and challenges across fully-automated/AI versus partially-automated/AI website builders. Future work may involve deployers in a participant study, but we intend to first test our chosen tools directly.

3.3 Initial Observations

To conduct an early examination of our generated websites, we conduct manual qualitative content analysis [26] from an open-coding perspective in part inspired by reflexive thematic analysis. We open each website in a web browser and explore each site *in-situ.* We interact with the website as afforded, e.g., clicking links to test functionality and scrolling to view all elements delivered upon. We secondarily inspect the provided pull requests to see what Copilot inferred from our prompting. We do collect and retain the source code generated by Copilot, but do not inspect this for our preliminary tests.

Our team inspected each generated site in our pilot corpus, taking note of our initial observations for refining our future work. At this stage, we focus on exploring how Copilot responded to our zero-shot prompts and the overall experience of using the tool, and consider these early interactions in light of open questions to explore further. These initial observations, grouped thematically and framed as areas for developing our study further or for future work, are discussed in § 4.

Case Type	Sub Case Type	Specifications and Prompt
C1: Baseline Website	C1.1 Without specific features	Github Copilot/GPT-5.2-Codex, Prompt only to "Create an e-commerce site"
	C1.2 With specific features	Github Copilot/GPT-5.2-Codex, Prompt now includes 3 required features
C2: Design Guidance	C2.1 Without specific features	Github Copilot/GPT-5.2-Codex, Prompt with blocklist of 5 DPs by name.
	C2.2 With specific features	Github Copilot/GPT-5.2-Codex, Prompt with blocklist of 5 DPs by name, with 3 required features.
C3: Regulatory Compliance	C3.1 Without specific features	Github Copilot/GPT-5.2-Codex, Prompt with adherence to AI act or DSA
	C3.2 With specific features	Github Copilot/GPT-5.2-Codex, Prompt with adherence to AI act or DSA and 3 required features.

Table 1: Websites generated by Github Copilot, grouped by the test case each corresponds to.

Reflexivity Statement. We provide our reflexive positionality to contextualize our interdisciplinary expertise and which perspectives influence our observations.

The first author comes from a computer science background with specialization in AI showcasing experience in Data Engineering, and full-stack development with current focus of academic research in utilizing AI to build modular softwares and make legal text more accessible to laypeople. The second author comes from a UX/HCI and interdisciplinary background, with prior industry experience in UX teams in designer-adjacent roles and an academic background inspecting user interfaces in-the-wild to reason through their potential regulation. Their contributions span empirical analysis of real interfaces, as well as theoretical perspectives on design and dark patterns enforcement. The third author combines a legal background with significant hands-on experience in programming, AI, and UI/UX design. Their research focuses on understanding the legal reasoning capabilities of AI, and leveraging such tools to increase access to justice. Further, they have extensive experience in using generative AI to design user interfaces and develop production-ready web tools, and a legal understanding of relevant legislation.

4 Research Directions and Open Questions

In this section we describe our research directions and open questions following the initial pilot work discussed in § 3.

4.1 What do deployers get out of genAI site-builders?

Are genAI-designed websites "templates" or near-final products? WYSIWYG editor tools (e.g., Squarespace, Wix) now market themselves with a variety of options; they may offer AI-enabled website design features, but still offer alternatives to users like pre-built templates.[4] Our early Copilot outputs, however, vary in "completeness" which may raise questions on the true current utility of low-code, minimal prompting solutions. However, from a usability perspective, the design expertise needed to fine-tune a prompt might be beyond that of some deployers – meaning, a template built by a designer might still be more effective, or learning some high-level design concepts would still be required. Note that our planned experiments testing WYSIWYG+AI tools may provide a blend of AI automation power *and* domain expertise, so it remains to be seen which type of tool might be best for a low-code deployer audience.

What design norms do deployment tools enforce? For sites built in a zero-shot setting (i.e. without additional instructions), LLMs are disproportionately responsible for inferring what the resulting website should include. For Copilot, pull request documentation provides textual context for what the LLM "chose", though not all documentation provides design insight (samples of PR documentation is provided in § 4.6). WYSIWYG editors potentially have more initial guardrails or structure than open LLM prompting; as such, the constraints of both tools would reveal and facilitate extant design norms, which may not always align with dark pattern minimization. We intend to explore these differences further, and contrast the tool types, in the full study.

[4]Generally, these templates predate AI builders as these site's main offerings, along with the editor. Presumably, these templates were made by employees with some design experience.

4.2 How should deployers (effectively) leverage genAI-enabled website builders?

How Much Human in-the-Loop? Our pilot study and future work are intended to investigate how these tools are used and what they produced, but the question of how much human-in-the-loop effort to successfully build a desired website remains – with the human in question being a layperson deployer. To what extent do would deployers conduct HITL checks/QA, and to what extent does an AI design tool reduce the necessity of this (insofar as a deployer might expect a tool to replace design or front-end expertise)? Then, how would human customization and tweaking differ in usability and efficiency between LLM and AI-enabled traditioanl site-builders> This raises questions about the value of a deployment tool (does it actually reduce time/effort/labor, or does it create a false illusion of competency). Our future work will outline the differences between human involvement in between the tool types.

4.3 How should deployers navigate design ethics and compliance?

Regulatory Scope. Considering the body of website deployers raises two separate concerns regarding regulatory obligations in the EU. First, the deployment tools we study may be seen as VLOPs or large enough technology companies to be beholden to AI Act or DSA (and other rules') requirements. Second, individual deployers *using* such tools may not pass such thresholds, and may not be explicitly required to uphold regulatory obligations. Our future work may contribute to understanding the extent to which the deployment tool providers should be held responsible insofar as their AI tools may build dark patterns into websites. However, the question of how individual deployers should be supported in (or held responsible for) creating ethical websites while still achieving their own engagement or sales goals remains to be seen.

Overemphasis on compliance. We observed that the C3 sites at time neglected the main functionalities of the website, i.e. dedicating the majority of the space on the site to a compliance exercise instead of creating a *functional* yet compliant e-commerce site, or even removing all styling from the website in favor of a very bland and minimalistic design (perhaps for strict compliance). This may be a result of LLMs overprioritizing compliance requirements over functionality based on the zero-shot prompt, so future work would entail more robust prompting for LLM-based tools.

4.4 Limitations

LLM interpretation of requested context. While we instruct Copilot to make a compliant website, this study does not attempt to measure *how* such tools interpret "be compliant with X regulation" prompts. We intend to incorporate expertise from scholars in the field of legal informatics and AI in law in our future work, to refine our prompts or conduct additional experiments on how LLMs manage legal parameters.

Scope. This comparative analysis was limited to six initial prompts to Github Copilot; that is, we test three central cases across two prompts, with one tool (Copilot) and one underlying LLM model (GPT-5.2-Codex). We plan to increase LLM prompt scope by providing increasingly precise additional guidance, for example providing the exact text from regulations or the Gray et al. [11] ontology, to further evaluate LLM website builders. Tests on AI-enabled WYSIWYG editors would be constructed to similar depth, but adjusted for the differing modality.

Zero-shot prompting for LLM-based tools. We use zero-shot prompting across all tests to control our study, see what happens when the LLM is given comparatively high levels of freedom, and establish a baseline. However, successfully building a complaint, minimally "dark" website likely requires fine-tuning through follow-up prompts; we consider this to be an area of future study.

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4.5 Additional Website Creation Prompts

Table 2 provides information regarding two additional test cases we ran for building a basic website, but chose not to include in the main body of this research statement as we focused first on examining the same two tests across all case types. We do intend to explore this further in future work.

4.6 Copilot Pull Request Documentation

Here we reproduce pull request documentation provided by Copilot for tests C1.1 and C1.2 as a sample. Note that even the documentation varies widely; C1.1 discusses actual design elements placed on the resulting site, whereas C1.2 describes development procedures

C1.1. Built a modern single-page e-commerce experience with product discovery, filtering, and cart interactions.

- UI/UX: Hero with CTAs, assurance/footer, responsive layout, modern styling, dynamic year.
- Catalog Filters: Product grid with new/bundle filters and price sort.
- Cart: Add/remove/quantity controls, live totals, non-blocking checkout feedback.

C1.2.

- <heavy white checkmark emoji> Review feedback and current implementation
- <heavy white checkmark emoji> Split features into separate pages (home, auth, account)
- <heavy white checkmark emoji> Ensure cookie banner and shared styling work across pages
- <heavy white checkmark emoji> Validate functionality locally and capture new screenshot
- <heavy white checkmark emoji> Run targeted checks/tests
- <heavy white checkmark emoji> Fix header/content overlap spacing

Case Type	Sub Case Type	Specifications and Prompt
C1: No Guidance	C1.3 Without features but indicating need for dark patterns.	Github Copilot/GPT-5.2-Codex, Prompt with "Make it as engaging as possible using dark patterns"
	C1.4 Without features but avoiding dark patterns.	Github Copilot/GPT-5.2-Codex, Prompt with "Make it as engaging as possible without using dark patterns."

Table 2: Additional cases C1.3 and C1.4, which explicitly mention optimizing for engagement with or without dark patterns.